

Chapter 1

Phonological profile changes in the Macro-Sudan Belt: Antagonism between ATR and interior vowels

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Rolle et al. (2020) present a fine-grained areal distribution of two vocalic properties in the Macro-Sudan Belt (MSB – Güldemann 2008): advanced tongue root (ATR) harmony and interior vowels (i.e. non-peripheral vowels [ɪ ʊ ɜ ə ʌ ...]). Their study determined that these two properties are antagonistic at both the level of individual languages (languages in the MSB tend to have either one or the other, rarely both), as well as at the areal level (ATR-harmony languages and interior-vowel languages are largely in complementary geographic distribution). The focus of this paper is examining diachronic phonological profile changes which take place as language families spread into and out of these geographic zones. We focus on two meso-areas: an East African ATR zone (centered around South Sudan), and a Central African ATR-deficient Zone, where phonemic and allophonic interior vowels are disproportionately prevalent. Specifically, we present two case studies illustrating phonological profile changes in order to comply with local areal patterns, and demonstrate how the aforementioned zones act as strong areal magnets.

1 Introduction

This paper focuses on the areal distribution of two vocalic features in the Macro-Sudan Belt (MSB; Güldemann 2003, 2008, 2010, 2018; Clements & Rialland 2008). One is Advanced Tongue Root (ATR) contrast and harmony, e.g. /i u e o/ vs. /ɪ ʊ ɜ ə ʌ/ contrasts. The other is interior vowels, i.e. non-peripheral parts of the vowel space, including non-low central vowels (e.g. ɪ ʊ ɜ ə, etc.), front round

vowels (e.g. y γ $\emptyset$ œ , etc.), back non-round vowels (e.g. u γ Δ , etc.). We show that ATR and interior vowel systems are in complementary areal distribution in the MSB, defining distinct meso-areas. We also show that these meso-areal signals are strong and stable, as they resist population and language movements: languages change their phonological profile when changing areas, rather than the other way around.

We first discuss the antagonistic distribution of ATR and interior vowel systems in the MSB in Section 2. We then show in Section 3 that languages change their phonological profiles when changing areas, focusing on two case studies: Sara-Bongo-Bagirmi (Central Sudanic) languages in Section 3.1 and the Bantu languages of the northeastern Congo Basin in Section 3.2. Finally, Section 4 discusses the few attested cases of vowel systems including both interior vowels and ATR contrast and harmony, and their potential interest for the understanding of the strong antagonism that characterizes these two features in the MSB.

2 ATR / interior antagonism

Our starting point is Rolle et al.'s (2020) ALFA vowel database (Areal Linguistic Features of Africa), which contains data from the vowel systems of 681 language varieties in the MSB, coded for phonemic contrasts and allophonic variants.¹ Our focus will be on two main variables. One is Advanced Tongue Root vs. Retracted Tongue Root contrast/harmony, and the other is the presence, number, and phonological role of interior vowels in the inventory.

2.1 ATR contrast and harmony

In canonical ATR systems, vowels are split into two mutually exclusive groups within a relevant phonological domain (e.g. the phonological word). In the [+ATR] group, a vowel canonically shows advancement of the tongue root, which widens the pharyngeal cavity, whereas [-ATR] vowels do not (cf. Stewart 1967, 1971; Casali 2003, 2008, 2016, among many others). Acoustically, [+ATR] vowels tend to have a lower first formant frequency (F1) than their [-ATR] counterparts. Since F1 is also the primary cue to contrasts in vowel height, [+ATR] vowels are often transcribed using a phone with a higher tongue body position compared to their [-ATR] counterparts. For example, Degema [deg] (Edoid: Nigeria, Kari 2007) is a language with a full set of ATR contrasts and active ATR harmony, as illustrated

¹The database is available in the online supplementary materials to Rolle et al.'s (2020): <https://www.degruyter.com/document/doi/10.1515/lingty-2019-0028/html>

in (1) below. The vowels of the possessive marker agree in [ATR] value with the noun it follows (tone is omitted).

- (1) Degema ATR contrast and harmony
[+ATR] vs. [-ATR]
- a. /ĩ ɛ ɔ ɔ ɔ ɔ/ (~ i e ɜ o u) /ĩ ɛ ɔ ɔ ɔ ɔ/ (~ ɪ ɛ a ɔ ʊ).
 - b. [+ATR] [ubi mee] ‘my palm kernel’
 - c. [-ATR] [ʊbɪ mɛɛ] ‘my book’

In many languages, the [+ATR] low vowel /a/ is missing, in which case the [-ATR] low vowel /a/ is often neutral. This is the case, for example, in nearby Kalabari [ijn] (Ijoid: Nigeria – Akinlabi 1997), where [+ATR] /i e o u/ contrast with [-ATR] /ɪ ɛ ɔ ʊ/, and /a/ is neutral with respect to ATR harmony.

Rolle et al. (2020) define two types of ATR systems: (i) complete systems, also known as Five-Height systems and characterized by cross-height harmony (e.g. Degema and Kalabari); and (ii) incomplete systems, also called Mid-Height harmony or Four-Height (M) systems, which lack [-ATR] high vowels /ɪ/ and /ʊ/. Standard Yoruba is a prototypical example of the latter type: mid-close vowels /e o/ do not co-occur with mid-open /ɛ ɔ/, but both are compatible with the high vowels /i u/, as shown in (2) (Przedziecki 2005:75-76).

- (2) Yoruba
- a. [oko] ‘farm’ (*okɔ)
[ètè] ‘lip’ (*etɛ)
 - b. [ɔko] ‘husband’ (*ɔko)
[ètè] ‘leprosy’ (*ete)
 - c. [ebi] ‘hunger’
[ilé] ‘house’
[ébí] ‘family, relations’
[ilè] ‘finger’

It is debatable whether the Incomplete type is really an instance of ATR contrast and harmony, since the patterns can easily be described without resorting to any other feature but vowel height: Standard Yoruba seems to involve mid-height co-occurrence restrictions rather than a full-blown ATR contrast and harmony.

Rolle et al.’s (2020) survey reveals the distribution of these ATR systems in the Macro-Sudan Belt shown in Figure 1. Complete ATR systems with cross-height harmony are found mainly in three areas (the dark blue dots): a small Atlantic

ATR zone, a larger West African ATR zone stretching from Côte d'Ivoire to Nigeria, and an East African ATR zone roughly centered on South Sudan and spilling over into neighboring countries to the north and south. This distribution also defines a Central African ATR-deficient Zone, wedged between the West and East African ATR zones, which is noticeably characterized by a total lack of complete ATR systems (the light blue dots) and systems entirely without ATR (gray dots).

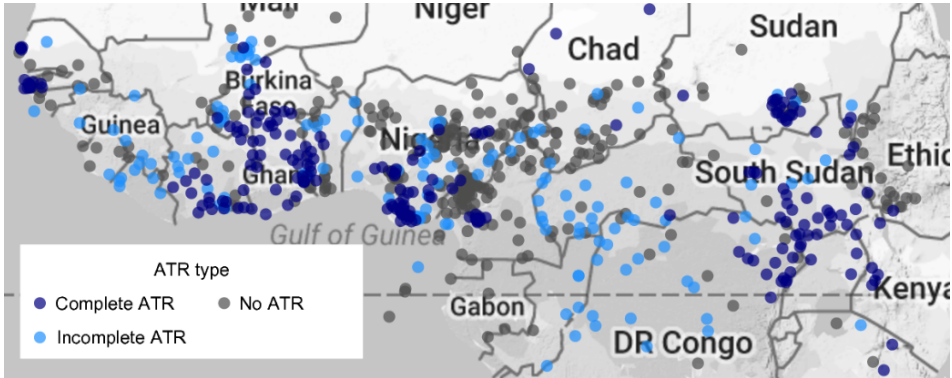


Figure 1: Areal distribution of ATR contrast and harmony in the Macro-Sudan Belt (Rolle et al. 2020:133)

2.2 Interior vowels

The next variable is the presence, status, and number of interior vowels in a language. As stated, interior vowels are defined negatively as all non-peripheral vowels, i.e. front rounded vowels (e.g. [y ø œ] etc.), back unrounded (e.g. [u ɤ ʌ] etc.), and non-low central vowels (e.g. [ɪ ʊ ə ɐ] etc.). Interior vowels are a well-known feature of many language families in Central Africa, in particular Bantoid and Chadic. For example, the Grassfields Bantu language Babanki [bbk] (Akumbu & Fogwe Chibaka 2012) boasts a four-way front/back contrast among high vowels /i ɪ ʊ u/ and a three-way contrast among mid vowels /e ə o/, as illustrated in (3) and (4) respectively.

- (3) Babanki high vowels
- a. /i/ tʃɪ ‘in-law’
 - b. /ɪ/ tʃɪ ‘fireplace’
 - c. /ʊ/ tʃʊ ‘spit’

d. /u/ kàntfù ‘cat’

(4) Babanki mid vowels

a. /e/ tʃê ‘minimize’

b. /ə/ tʃó ‘kick’

c. /o/ tʃô ‘pass’

Interiority may manifest both as bona fide phonemes as well as allophonic variants of peripheral vowels. Dialectal variants of Ibibio [ibb] (Delta Cross), for example, differ in whether interior vowels [i ʊ ə ʌ] are phonemic or not. At the surface level, however, all occur in ‘General Ibibio’ as conditioned variants of underlying /i u o/, shown in (5) (Urua 2000:30).

(5) Ibibio interior allophones

a. /kím/ ‘sew’ [kím]~[kám]

b. /ùkù/ ‘fox-like animal’ [ùkù]~[ùkù]

c. /kpók/ ‘cut into pieces’ [kpók]~[kpák]

Finally, there are languages where the only interior vowel is /ə/ (variously transcribed as [ə], [ɜ], [ɐ]), i.e. the [+ATR] counterpart of the [-ATR] low vowel /a/. This was the case in Degema in (1) above. This vowel can be considered peripheral within its own class of [+ATR] vowels, and qualifies as interior only from the point of view of its phonetic realization. In Rolle et al. (2020), such vowels were specially coded.

The geographic distribution of the different analytic levels of interiority is shown in Figure 2. Languages with phonemic interior vowels are in red, languages whose interior vowels are non-phonemic (i.e. conditioned variants of peripheral vowels) are in orange, and languages whose only interior vowel is the [+ATR] counterpart of the [-ATR] low vowel /a/ are in purple. Languages entirely lacking interior vowels are in gray.

The largest concentration of phonemic interior vowels (in red) is found in Central Africa, in what we term the Central African Interior Vowel Zone. Languages of this area almost systematically have phonemic interior vowels. This zone encompasses languages of southern Chad moving into Nigeria and Cameroon. This area has one of the highest concentrations of language varieties in Africa, and is at the confluence of three of the four Greenbergian phyla. This includes language families such as Grassfields Bantu, Delta-Cross, Adamawa, Ubangian, Kainji, Platoid, Jukunoid, all of Chadic (West, Biu-Mandara, Masa, and East Chadic), and Sara-Bongo-Bagirmi.

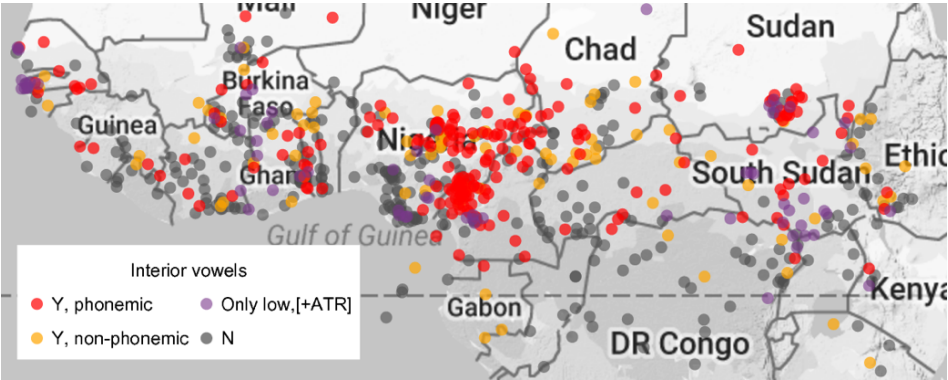


Figure 2: Areal distribution of interior vowels in the Macro-Sudan Belt (Rolle et al. 2020:142).

2.3 ATR / interiority antagonism

While cases of co-occurrence of ATR and interior vowels within the same vowel system are attested, they are the exception rather than the rule. One such system is illustrated by Kanembu [kbl], whose vowel inventory is shown in Table 1 (Saharan: Chad – Jouannet 1982, whose transcription conventions for [ATR] we reproduce here).

Table 1: Kanembu vowel inventory (Jouannet 1982)

[ATR]	Front	Central	Back
+	i	ɨ	u
–	ɪ	ʉ	ʊ
+	e	ə	o
–	ɛ	ʌ	ɔ
Ø		a	

However, such systems are very rare. In fact, one of the key findings from Rolle et al.’s (2020) study is the complementary geographic distribution between ATR systems and Interior systems. This is shown in Figure 3. This map clearly shows that ATR and interior systems do not frequently co-occur: purple dots (representing languages with both) are much rarer than blue (ATR only) and red (interior only) dots.²

²Rolle et al. (2020) propose two different types of maps: a ‘liberal’ map in which the ATR and

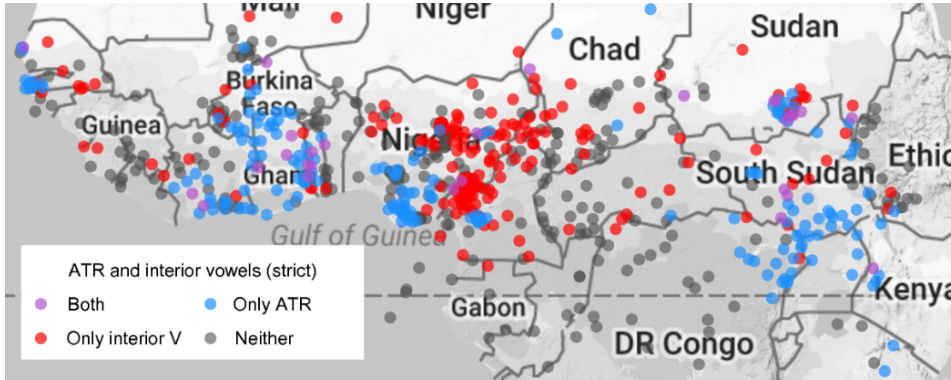


Figure 3: Distribution of ATR and interiority.

Furthermore, ATR and Interior systems appear to be in complementary geographic distribution, shown in Figure 4. This figure maps out five meso-areas defined within the MSB on the basis of the distribution of the two criteria of ATR contrasts and harmony and interior vowels. Most of the interior systems cluster in a Central African Interior Vowel Zone (the dashed line), which exists entirely within the Central African ATR-deficient Zone (meso-area 4).

As we said above, the ATR contrast is partly cued by F1, i.e. realized with a distinction that is perceptually akin to vowel height, on the “vertical” axis of the vowel space. In contrast, interiority is mostly cued by F2 and adds additional contrasts along the backness/horizontal dimension. This antagonistic relationship therefore seems to be rooted in articulatory and perceptual pressure on what shapes vowel inventories. It seems like in the MSB, languages tend to expand their vowel system either vertically through an ATR contrast, or horizontally through interior vowels, but rarely both, and that the “choice”, so to speak, is mostly determined by the area where the language is spoken.

3 Changing profiles when changing areas

The question that we would like to address now is the following: how strong are these meso-areal signals? Are they old and stable (e.g. persisting through population and language movements), or recent and unstable?

interior categories include all cases of ATR (complete and incomplete) and interior systems (phonemic, allophonic and [+ATR] version of /a/), and a ‘strict’ map which includes only Complete ATR systems and systems with phonemic interior vowels. We show only the ‘strict’ map here (cf. Rolle et al. 2020: 146-147).

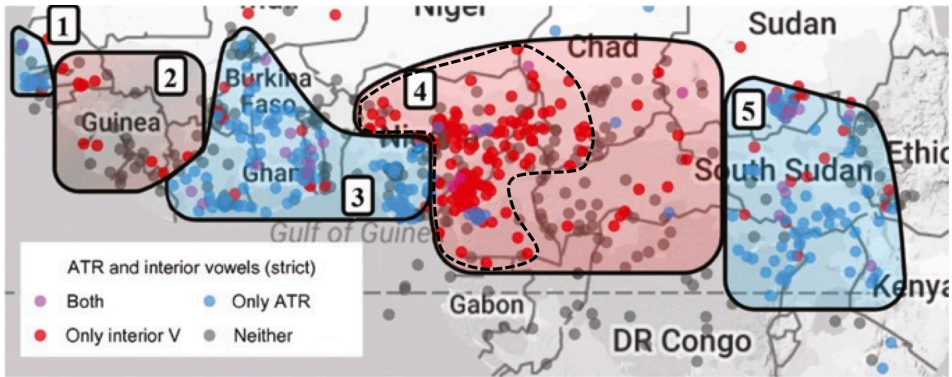


Figure 4: Five areal zones defined by vocalic systems in the Macro-Sudan Belt: [1] Atlantic ATR zone, [2] Guinean ATR-deficient Zone, [3] West African ATR zone, [4] Central African ATR-deficient Zone (includes Central African Interior Vowel Zone in dashed contour), [5] East African ATR zone.

Our current findings suggest a high stability of these areal signals, seen in particular by the fact that languages adapt their phonological profile to the areas where they are situated or where their speakers migrate to. In other words, the ATR and interior vowel meso-areas defined above are characterized by areal features that survive and “percolate” through layer after layer of population movements and language changes, a case of “vertical diffusion through time” discussed by Güldemann (2010:576); see also Güldemann & Fehn (2017) and Güldemann (2018:497–498). In this section, we present two case studies in support: Central Sudanic languages (specifically the Sara-Bongo-Bagirmi family), and the Bantu languages of the northeastern Congo Basin.

3.1 Central Sudanic languages

Central Sudanic languages are spoken along an arc stretching from the northeastern Congo basin to southern and central Chad, through western South Sudan and the northern fringes of the Central African Republic (as seen on the map in Figure 5). Notice that we overlay on this map where the approximate boundaries are for the Central African ATR-deficient Zone and the East African ATR Zone. Within the latter, four Central Sudanic language families out of a total of six are spoken in or near the northeastern quadrant of the Democratic Republic of the Congo – Moru-Madi, Mangbutu-Efe, Mangbetu, and Lendu – making it the region where the highest level of diversity is to be found within the family.

Proto-Central-Sudanic is thus likely to have been spoken in or near this area (according to the ‘center of gravity model’ proposed by Sapir 1958 [1916]: 455-458) – a hypothesis explicitly made by Ehret (1974).

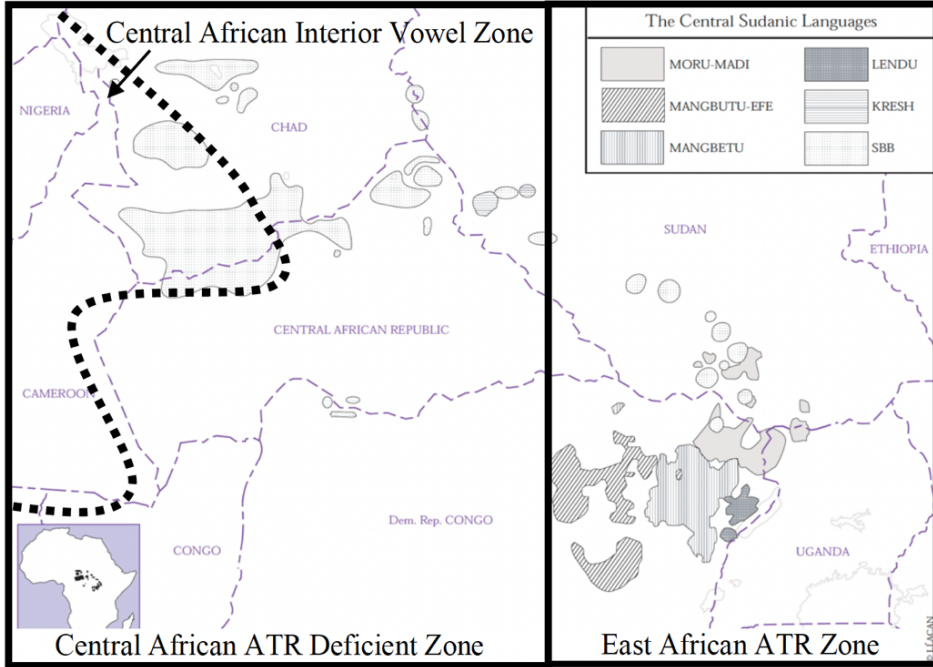


Figure 5: Map of Central Sudanic languages with respect to the distribution of ATR and Interior Zones (adapted from Boyeldieu 2006b)

Within Central Sudanic, the Sara-Bongo-Bagirmi branch is of particular interest for our purpose (the more northerly-located areas of Central Sudanic, labeled ‘SBB’ in Figure 5). The internal history of this SBB family strongly suggests that Proto-SBB was spoken in the east of present-day South Sudan (Boyeldieu 2000: 166–175, 2006b, 2009), and that SBB speakers subsequently migrated in a north-westerly direction, all the way to southern and central Chad. A map showing the SBB expansion is shown in Figure 6. As can be seen, the Proto-SBB homeland is in the East African ATR zone, and most of the language family then expanded outside of this zone, into the Central African ATR-deficient Zone. Later still, there was an expansion all the way into the Central African Interior Vowel Zone, where some present-day SBB languages are spoken.

This geographical distribution correlates with the vowel system profiles of the languages. The SBB languages still spoken in the East African ATR zone

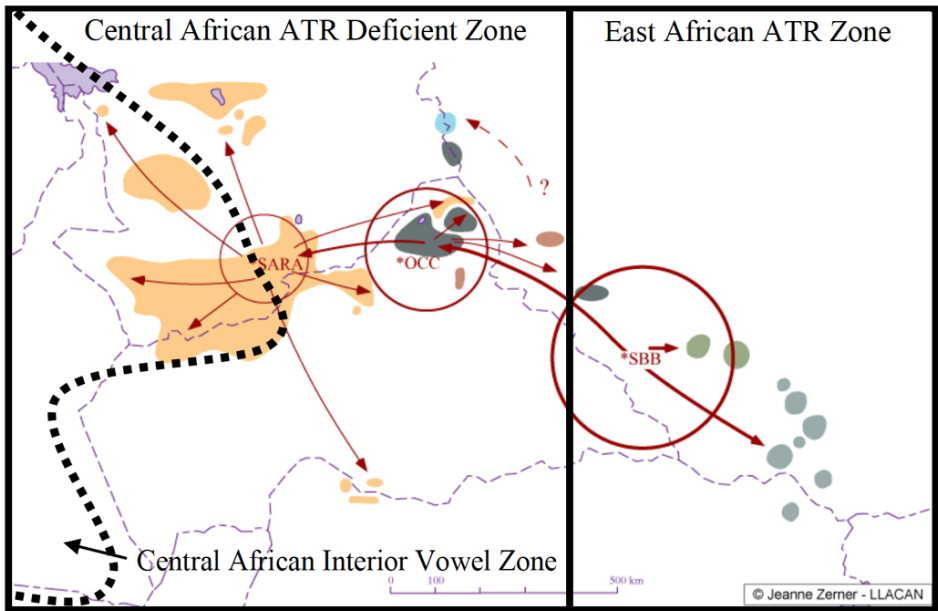


Figure 6: Map of the Sara-Bongo-Bagirmi expansion (adapted from Boyeldieu 2006a)

all have an ATR contrast and ATR harmony, e.g. Bongo, Baka, Jur Mödö, 'Bëli, and Morokodo, whose vowel inventories are shown in Table 2 (vowels in square brackets [...] are allophones with no phonemic status).

Table 2: Vowel systems of non-Western SBB languages (data from Persson 2004)

(a) Jur Mödö [bex]			(b) 'Bëli [blm]			(c) Morokodo [mge]			(d) Bongo [bot]			(e) Baka [bdh]		
i		u	i		u	i		u	i	[ɯ]	u	i	i	u
ɪ		ʊ	ɪ		ʊ	ɪ		ʊ	ɪ		ʊ	ɪ		ʊ
e	ə	ə	e	[ə]	ə	e		ə	e		o	[e]	[ə]	[ə]
ɛ		ɔ	ɛ		ɔ	ɛ		ɔ	ɛ		ɔ	ɛ		ɔ
a			a			a			a			a		

ATR harmony is illustrated in the following Jur Mödö example, whereby the ATR value of a noun spreads to the immediately preceding head noun in an inalienable possessive construction – here the head noun is dɔ ~ də ‘head’ (/ə/ being the [+ATR] counterpart of /ɔ/).

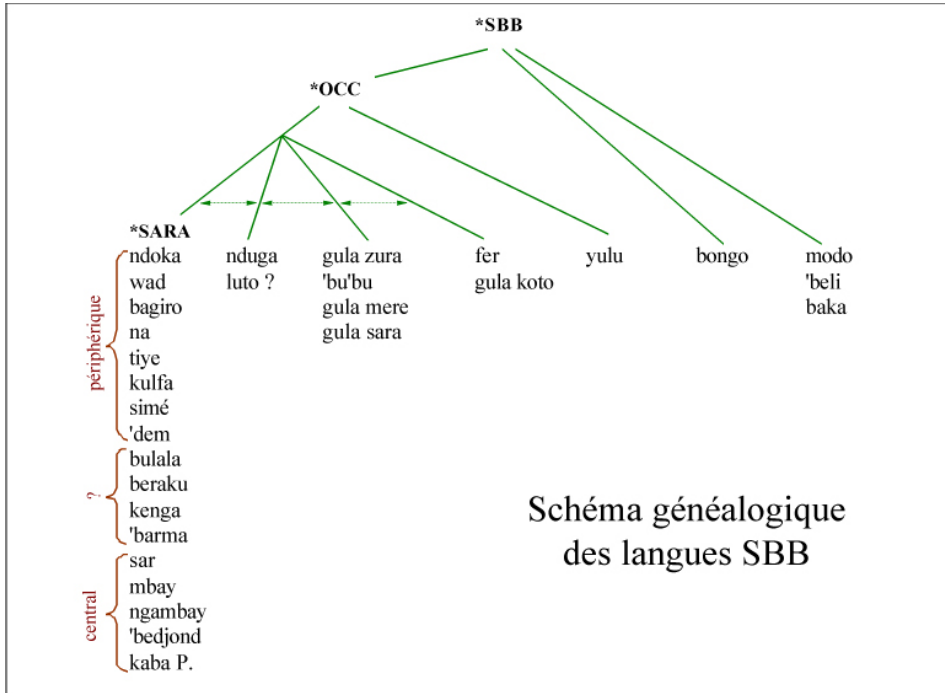


Figure 7: Classification of SBB languages (from Boyeldieu 2006a)

Jur Mödö ATR harmony (Persson & Persson 1991:7; Persson 2004:80)

- (6) a. *dò ɲòkɔ́* ‘head of a wild cat’
 b. *dò kèdè* ‘head of an elephant’

None of the SBB languages spoken outside of the East African ATR zone have the ATR contrast. These constitute the majority of SBB languages, and all belong to the Western branch of the family (referred to as *OCC in Figure 7). This is illustrated in Table 3 with the vowel systems of five languages – one per terminal branch in the family tree in Figure 7, going from the westernmost (Ngambay) to the easternmost (Yulu).³

³Sources: Vandame 1963 and Nangone et al. 2015 (Ngambay), Palayer 2006 (Deme), Olson 2013 (Lutos), Nougayrol 1999 (Gula), Boyeldieu (1987) (Fer and Yulu). Nougayrol (1999:51) transcribes the two central vowels attested in (some) Gula varieties as <ə> and <ʌ>, but describes them as close and close-mid respectively. Accordingly, we have taken the liberty to transcribe them [i] and [ə] respectively. Olson (2013) also includes the diphthong /ʊa/ in the inventory, but its

Table 3: Vowel systems of Western SBB languages

a. Ngambay	b. Deme	c. Lutos	d. Gula	e. Fer & Yulu
i [i] u	i u	i u	i [i] u	i u
e [ə] o	e o	e (ə) o	e (ə) o	e [ə] o
ɔ	ɛ ɔ	ɛ ɔ	ɛ ɔ	ɛ ɔ
a	a	a	a	a

Because Proto-SBB was most likely spoken in the East African ATR Zone, one can hypothesize that it had an ATR contrast and ATR harmony.⁴ This is maintained in the SBB languages that are still spoken in the same zone. The SBB languages that migrated out of the East African ATR Zone all lost ATR.

Interestingly, the westernmost Sara languages (Boyeldieu’s (2006a) “central Sara” in Figure 7) all have at least one interior vowel which, most of the time, acts as a reduced vowel in certain phonotactic environments, viz. the majority of disyllabic words either have two identical vowels, or one of them is [i] (Keegan 1996). This reduced vowel is extremely frequent both in the lexicon and in daily speech. In other words, despite the fact that the interior vowel is most of the time non-phonemic, vowel interiority is extremely salient in speech. Some languages have more than one interior vowel, e.g. Ngambay also has the mid-vowel [ə], a contextual allophone of /e/. It is tempting to see the presence of interior vowels in these central Sara languages as areal alignment resulting from their integration into the Central African Interior Vowel Zone (as seen on the map in Figure 6). Indeed, these languages are in contact with languages of other families that have contrastive interior vowels: Chadic languages like Tumak (Caprile 1975, 1977), or Sibine (Barreteau & Jungraithmayr 1982); Riverine Bua languages such as Tun (Palayer 1975a), Kwaa Cinyi, Kwaa Perim, Kwaa Waany (Boyeldieu & Seignobos 1975), Lua (Boyeldieu 1985), as well as Bua languages formerly spoken before the arrival of SBB languages in southern Chad by populations now assimilated to SBB-speaking groups, e.g. Lɔɔ (Palayer 1975b). These languages, together with central Sara, constitute the easternmost extension of the Central African Interior Vowel Zone.

status is unclear. He adds that /ə/ is rare in the language, and very tentatively gives it phonemic status, although this is not clearly established.

⁴Boyeldieu (2006a) remains agnostic as to whether Proto-SBB is to be reconstructed with ATR, but does not consider it an unlikely hypothesis (pers. comm. 2018).

Two points suggest that the areal alignment hypothesis might be correct. First, the interior vowels are different in nature, role, and frequency in central Sara vs. SBB languages in the East African ATR Zone. In the latter, interior vowels are often only the [+ATR] counterparts of [-ATR] peripheral vowels (ɔ~ə in Jur Mödö, 'Bëli, and Morokodo, a~ə in 'Bëli and Baka); they can thus be considered to be peripheral vowels (and not interior) in the [+ATR] vowel subsystem. This leaves only the allophone [ɯ] of /u/ in Bongo, and contrastive /i/ in Baka, which do not have the widespread distribution of reduced [ɪ] in Sara languages. Interior vowels thus have different origins and functions in the easternmost vs. westernmost SBB languages.

The second argument in favor of the areal alignment hypothesis is that only the westernmost central Sara languages have interior vowels, i.e. those that are the closest to the Central African Interior Vowel Zone, in direct contact with languages with interior vowels. The Sara languages subsumed under the "outer" (*périphérique*) label in Boyeldieu's classification in Figure 7, spoken further east (in the portion of the Sara language area that is outside of the Central African Interior Vowel Zone in Figure 6) only have peripheral vowels, and lack the reduction process at work further west, as illustrated with Deme in Table 3. Outer Sara languages are mostly in contact with languages without interior vowels, notably Inland Bua languages (Fanya, Kulaal, Zan Gula) – although a few neighboring languages also have interior vowels, e.g. Sar (SBB, Palayer 1992:17–18) and Lua (Bua; Boyeldieu 1985:136) in contact with Deme, or Aiki (Nougayrol 1990:21–23) in contact with Na.⁵ This distribution is suggestive of the innovative character of interior vowels in central Sara. Boyeldieu (2006a), in his preliminary reconstruction of the Proto-SBB vowel system, does not include interior vowels, and states that the reduced interior vowels of central Sara languages are indeed innovative. He does not say anything about the interior vowels attested in the SBB languages spoken in the East African ATR Zone, but his proposed reconstructions suggest that they might also be innovative.

On the basis of the facts presented above, we hypothesize that central Sara languages developed interior vowels under the influence of the languages they came into contact with when they entered the Central African Interior Vowel Zone – i.e. through areal alignment. In conclusion, not only did the SBB languages which migrated into the Central African ATR-deficient Zone lose their ATR contrast, those that went all the way into the Central African Interior Vowel

⁵Although the intensity of the potential contacts between Na and Aiki speakers is not known, it can be hypothesized to have been rather shallow, and limited to the easternmost fringe of the Na-speaking area, given the low population density characterizing the Dar Runga where Aiki is spoken (Nougayrol 1990:9–15).

Zone developed high-frequency interior vowels. This constitutes a striking case of areally determined phonological profile change.

3.2 Bantu languages of the northeastern Congo Basin

The ALFA database includes 44 Bantu languages, all spoken in or near the Macro-Sudan Belt, i.e. in the northern fringes of the Bantu spread zone. Only 11 of these have ATR harmony, nine of which are spoken in the far northeastern quadrant of the Democratic Republic of the Congo and across the border in South Sudan and Uganda: Amba, Bila [Guthrie code D311], Budu [D332], Gungu [JE101], Konzo [JD41], Lika [D201], Nande [JD42], Talinga-Bwisi [JE102], and Vanuma [D331] (Kutsch Lojenga 2009 and references therein). The other two Bantu languages are Langi and Malila, which are spoken in Tanzania and at a considerable distance from these former nine languages (Kutsch Lojenga 2009). All other Bantu languages in the ALFA database have either no ATR contrasts, or a pattern falling under “incomplete” ATR as defined above.⁶

ATR harmony in Lika is illustrated in (7) below with co-occurring noun class prefixes and suffixes, which agree with the ATR value of the noun root (vowel inventory: [+ATR] /i e o u/, [-ATR] /ɪ ɛ a ɔ ʊ/; Kutsch Lojenga 2008).

(7) Lika ATR harmony

- a. Class 15
 - i. [-ATR] root: *kʊ-bɔbɔ-kʊ* ‘deceit’
 - ii. [+ATR] root: *ku-li-ko* ‘knee’
- b. Class 19
 - i. [-ATR] root: *(s)ɪ-kpɔkpɔ-sɔ* ‘forehead, skull’
 - ii. [+ATR] root: *(s)i-bokú-so* ‘skin, bark’

ATR harmony is notably absent from other Bantu languages, and it is not reconstructed in Proto-Bantu (Meeussen 1967; Schadeberg 2003; Hyman 1999; Bostoen 2019; a.o.). The Proto-Bantu vowel system is reconstructed with seven peripheral vowels, organized along four degrees of height. The contrastive feature(s) accounting for the four vowel heights are still debated. Four main hypotheses exist (cf. Schadeberg 2003; Bostoen 2019; and references therein). A tradition initiated by Meinhof (1899:18) and followed by Meinhof & Van Warmelo (1932:33),

⁶Casali (2003) includes 24 Bantu languages with ATR contrasts in his typology. Many of these correspond to incomplete ATR systems in our terminology, i.e. systems without ATR (cross-height) harmony.

Meeussen (1967:83), and Guthrie (1967:52), reconstructs a purely height-based contrast with typologically rare “superclose” vowels characterized by a very narrow degree of constriction, as seen in Table 4(a). Greenberg (1951) proposes the cross-linguistically more frequent four-height system in Table 4(b). Bastin et al. (2002) reinterpret the contrast between the highest two degrees as one of laxness/tenseness, i.e. as involving the feature [ATR], as shown in Table 4(c).⁷ Finally, Maddieson (2003:18–20) rejects the latter ATR-based analysis of the Proto-Bantu vowel system in favor of an interpretation of Meinhof’s “superclose” vowels *ɨ and *ʉ as fricative vowels *ɨ̥ and *ʉ̥ (Table 4(d)).⁸

Table 4: Reconstructions of Proto-Bantu vowel inventory

a.	Meeussen 1967	b.	Greenberg 1951	c.	Bastin et al. 2002	d.	Maddieson 2003
	ɨ ʉ		i u		i u		ɨ ʉ
	i u		e o		ɪ ʊ		i u
	e o		ɛ ɔ		e o		e o
	a		a		a		a

Space does not permit a detailed discussion of the question of the phonological characterization and phonetic realization of Proto-Bantu vowels. It is sufficient to merely point out that no ATR harmony is reconstructed for Proto-Bantu, and the existence of an ATR phonetic distinction and phonological contrast among high vowels is at best controversial.

Furthermore, Proto-Bantu is thought to have been spoken somewhere along the present-day border between Cameroon and Nigeria (cf. Grollemund et al. 2015, and references therein), i.e., in what corresponds today both to the Central African ATR-deficient Zone and the Central African Interior Vowel Zone. It is unclear whether these linguistic features had the same areal distribution at the

⁷The /i e ɛ a ɔ o u/ system proposed by Greenberg 1951 is also often described as involving an ATR contrast among the mid vowels (cf. Hyman 1999 and references therein).

⁸Maddieson’s main arguments are that (i) Proto-Bantu did not have ATR harmony, as shown by ATR-disharmonic reconstructed forms such as *kúnj ‘firewood’ and *kúpi ‘be short’; and (ii) the widespread spirantization caused by the superclose vowels in many Bantu languages is highly suggestive of a high degree of constriction of these vowels rather than a contrast in terms of advancement or retraction of the tongue root. Connell (2000, 2007) and Faytak & Merrill (2015) bring evidence in favor of this analysis by showing that some of the quirky aspects of Bantu spirantization are similar to what can be observed in Bantoid languages that have fricative vowels, and are thus more naturally explained if the Proto-Bantu “superclose” vowels were characterized by a degree of constriction such that they were realized with frication (cf. also Zoll 1995, who analyzes Bantu “superclose” vowels as involving a [+consonantal] feature).

time (some 5,000 years ago, cf. Vansina 1995:52, Blench 2006:126). Consequently, little can be hypothesized about the degree of areal alignment of the Proto-Bantu vowel system.

Regardless, we hypothesize that those Bantu languages which currently possess ATR harmony are innovative rather than conservative. As Bantu-speaking groups migrated into the northeastern DRC (i.e., into the East African ATR zone), they changed their phonological profile by adopting the ATR contrast and harmony patterns of the Central Sudanic languages they encountered.⁹ Indeed, all the Central Sudanic languages directly in contact with these Bantu languages in the northeastern quadrant of the Democratic Republic of the Congo have ATR contrasts and harmony, with vowel systems comprising four [+ATR] vowels /i e o u/ and five [-ATR] vowels /ɪ ɛ a ɔ ʊ/ – e.g. Mangbetu (Démolin 1992), or the Mangbutu-Efe languages Lese, Mamvu, and Efe (Vorbichler 1965, 1971, 1974). Active ATR harmony is illustrated with Mangbetu in (8), where one can see that the nominal prefix *nɛ-/ne-* and the imperative prefix *kʊ-/kù-* both agree with the ATR value of the root to which they are added.

(8) Mangbetu ATR harmony (Démolin 1992:86)

- a. *né-ŋò* ‘eye’ vs. *né-ŋrù* ‘head’
nè-kí ‘tooth’ vs. *nè-kí* ‘intestine’
- b. *kò-ò* ‘give!’ vs. *kù-ò* ‘go!’
kò-ndà ‘search!’ vs. *kù-ṭè* ‘bury!’

Note that the ATR feature in Northeast Bantu languages does not seem to be inherited from a common mid- or low-level ancestor, since the presence of ATR harmony cuts across genealogical subgroupings – at least according to Grollemund et al.’s (2015) phylogenetic classification (cf. Figure 8). ATR harmony is found in both the Central-Western branch (in particular languages from Guthrie’s group D) as well as Eastern branch (notably in Guthrie’s groups JE and JD), but importantly, only in those languages from these branches that are spoken in the East African ATR zone. Grollemund et al. (2015) propose a model for the Bantu migrations showing that Bantu languages in that region result from several independent population and language movements, as shown in Figure 9.¹⁰ Bantu languages moving into the East African ATR Zone thus seem to have adapted

⁹This hypothesis can be found in previous literature (e.g., Kutsch Lojenga 2008 about the origin of ATR harmony in Lika), although it has never been fully explored.

¹⁰The exact paths followed by the Bantu migrations are still debated (cf. Pakendorf et al. 2011, Grollemund et al. 2015, and references therein for more information).

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their phonological system to the local areal signal by innovating an ATR contrast and ATR harmony on an individual basis.¹¹

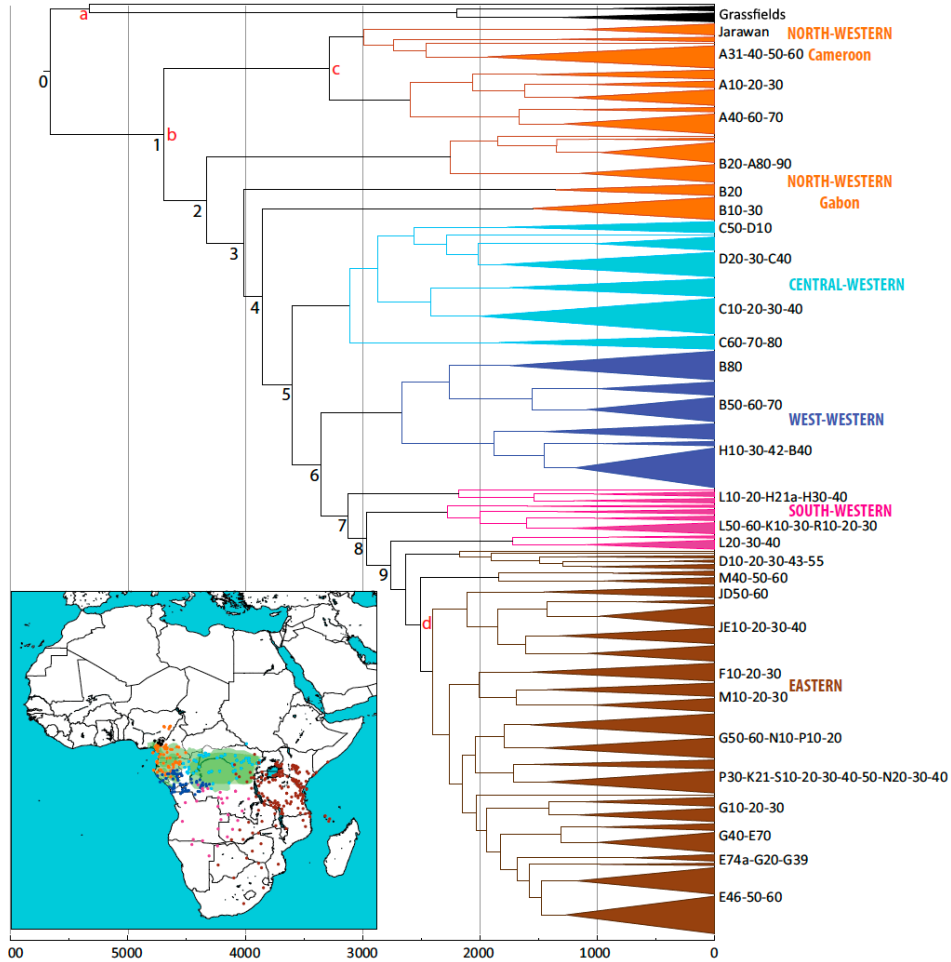


Figure 8: Grollemund et al.'s (2015) phylogenetic classification of the Bantu languages

¹¹Another feature that has been hypothesized to be acquired through areal pressure in these languages is labial-velar consonants (Bostoen & Donzo Bunza Yugia 2013; Idiatov & Van de Velde 2021).

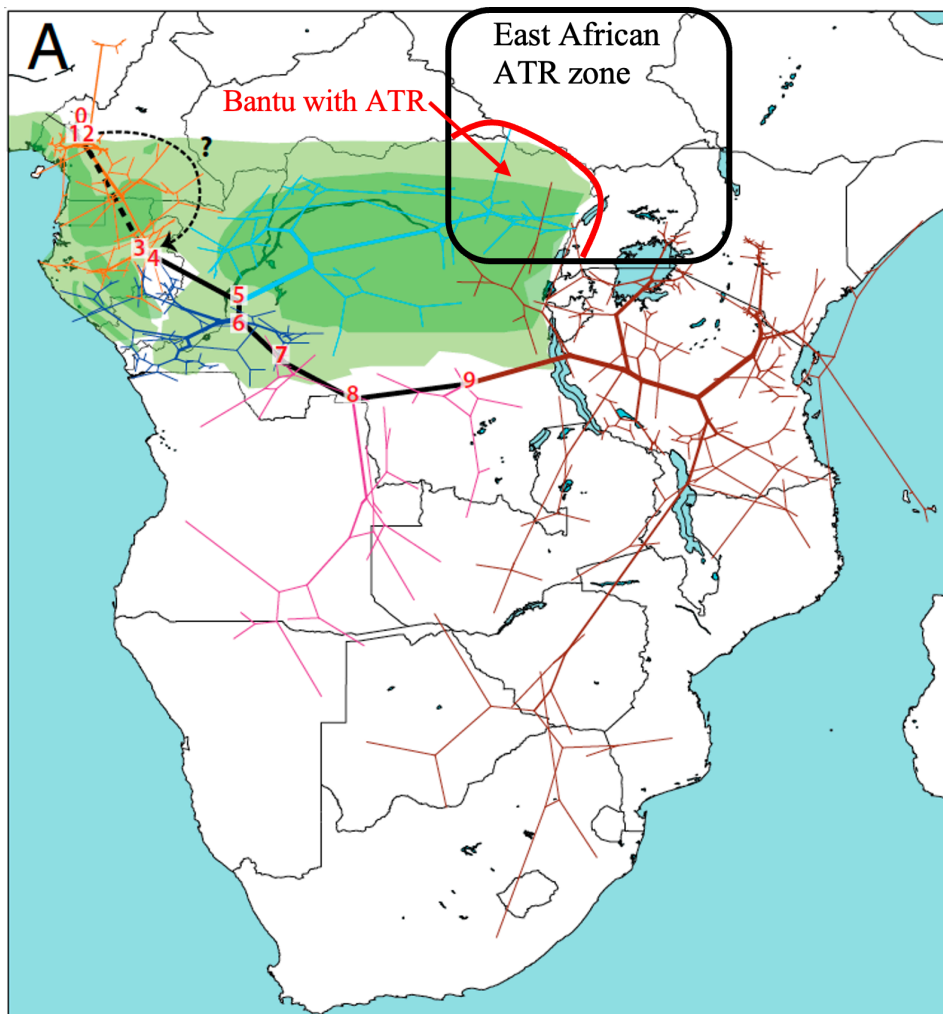


Figure 9: Hypothesized Bantu migration routes and location of the East African ATR zone and the present-day Bantu languages with ATR contrast and harmony (adapted from Grollemund et al. 2015)

4 ATR/interiority overlap

Let us briefly return to the complementary geographical distribution between ATR and Interior vowels in the MSB. As stated already, despite their general complementarity, languages with both features are readily attested, although they are not frequent. Some examples of languages with both phonemic ATR and interior vowel contrasts were found in Figure 3 (the purple dots).

To further understand the languages of the MSB which have both interior vowels and an ATR contrast, let us divide them into three groups. This is shown in Table 5. The first group consists of ATR languages whose interior vowels are only allophonic. This is the case of Nawuri, for instance, where all front vowels have a centralized allophone of the same height, i.e. /i ɪ e ε/ = [i ɪ e ε] ~ [i̠, i̠, ə, ɜ],¹² and /a/ has a [+ATR] allophone (Casali 1988).

The second type consists of languages with one or more phonemic interior vowels with no counterpart in ATR harmony (in some of these cases the interior vowel is neutral with respect to harmony). Such is the case of Anii, for example, which has 11 contrastive vowels, including the 10 symmetrical ATR system /i e ə o u/ vs. /ɪ ɛ ɔ ʊ/, as well as /i̠/, a [-ATR] vowel with no [+ATR] counterpart (Morton 2012).

The third type consists of languages with one or more series of phonemic interior vowels which take part in the ATR harmony in the same way as peripheral vowels, such as Kanembu (Table 1 above), which pairs [+ATR] /ə/ with [-ATR] /ʌ/.

As seen in the maps throughout, most of the languages with both ATR harmony and phonemic interior vowels are found in the West and East African ATR zones (zones 3 and 5 in Figure 4). It is surprising that so few are found in Central Africa (zone 4), especially in the Central African Interior Vowel Zone or at its boundaries. Indeed, one might have expected such systems to emerge (or be more stable) precisely where both areal signals meet. This does not appear to be the case. This suggests that ATR contrast and harmony is not as strong an obstacle to the development of interior vowels as phonemic interior vowels are to the development of an ATR contrast. This asymmetry must for now remain a tentative hypothesis. Indeed, languages with both ATR and interior vowels are so rare in the MSB that the skewed areal distribution could very well be due to chance. Note that the rarity of such saturated vowel systems is not cross-linguistically unexpected: only 184 languages in Maddieson's (2013) sample of 564 languages, i.e.

¹²We have taken the liberty to change Casali's (1988) original [i̠, i̠, ə, ʌ] to [i̠, i̠, ə, ɜ], a transcription which is more transparent.

Table 5: Languages with ATR and Interior vowels

	ATR+interiority system	Areal zone	Languages
a.	Allophonic interior vowel(s)	3	Gonja, Delo, Tampulma, Izi, Abe, Nawuri
		4	Waja, Mundang
		5	Pokoot, Ngiti, Olubo, Bongo
b.	Interior vowel(s) with no [±ATR] counterpart	3	Abron, Sekpele, Anii, Dagbani, Akebu
		4	Hone, Yukuben, Fur
		5	Baka, Burun
c.	Interior series with [±ATR] pair	1	Lehar
		3	Bete, Kouya, Godie, Lama
		4	Iceve-Maci, Kanembu
		5	Dagik, Tima, Tocho, Tira, Acheron
d.	Other (cf. §3.1)	5	'Bëli, Jur Mödö, Morokodo

33%, have at least 7 contrastive vowel qualities, and only three are listed as having more than 13 (British English, Bete, and German). This is most likely one of the factors explaining the antagonism between ATR and interiority in the MSB.

5 Conclusion

In conclusion, we have shown in this paper that vowel systems in the MSB have a clear meso-areal distribution characterized by an antagonism between ATR contrast and harmony and interior vowels. Central Africa is characterized by a conspicuous lack of ATR languages, and includes a wide and dense area where languages predominantly have interior vowels – making interior vowels another Central African feature (along with the labial flap, *inter alia*).

We have shown that languages adapt to the phonological profile of the area where they are spoken, or into which their speakers migrate. Such areal alignment took place in the history of Western Sara-Bongo-Bagirmi languages (a Central Sudanic branch), which lost ATR contrasts and harmony when migrating from the East African ATR zone into the Central African ATR-deficient Zone,

and expanded their interior vowels as they got closer to the Central African Interior Vowel Zone. We have also shown that the Bantu languages that entered the southwestern quadrant of the East African ATR zone developed ATR contrasts and a fully functional ATR harmony system not found in Proto-Bantu. The areal signals characterizing the meso-areas within the MSB are thus strong and stable: they resist migration and language shift, and percolate through multiple layers of population movements.

The results presented here are promising, but still limited. In particular, the factors at work in the antagonism between ATR and interior vowels in the MSB have yet to be fully understood. The ultimate origin of these two areal signals is also still mysterious. One may wonder what role the arrival of Chadic languages in Central Africa may have played in the creation of the Central African Interior Vowel Zone (interior vowels are prominent in virtually all Central Chadic languages, and many languages in other branches of the Chadic family as well, cf. Wolff 2022). Finally, the exact mechanisms at work in phonological profile changes have yet to be fully worked out. These are likely to be different from language to language, and their understanding requires detailed low-level comparative work, which is still lacking for many languages and language families in this part of Africa.

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